

Functional & Performance Testing Template

Model Performance Test

Date	17 June 2026
Team ID	-----
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	5 Marks

Test Scenarios & Results

Test Case ID	Scenario (What to Test)	Test Steps (How to Test)	Expected Result	Actual Result	Pass / Fail
FT-01	Weather & Rainfall Input Validation	Enter valid and invalid rainfall, temperature, humidity, and weather parameter values.	Valid inputs are accepted; invalid or empty inputs display appropriate validation messages.	System validates inputs correctly and displays error messages for invalid values.	Pass
FT-02	Numerical Input Validation	Enter values within and outside the acceptable range for rainfall and weather parameters.	Accepts valid numerical values and rejects out-of-range values.	Validation performed successfully.	Pass
FT-03	Flood Prediction Generation	Enter complete weather parameters and click Predict Flood Risk .	The system predicts Flood Risk or No Flood Risk using the trained XGBoost model.	Prediction generated successfully with correct output.	Pass
FT-04	Model Loading & Prediction Service	Launch the Flask application and verify that the saved	Trained model loads successfully and responds to	Model loaded successfully without errors.	Pass

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PT-01	Prediction Response Time	XGBoost model loads correctly.	prediction requests.		
		Measure the time taken to generate flood prediction after user input.	Prediction response time should be less than 2 seconds .	Average response time: 1.4 seconds .	Pass
PT-02	Concurrent Prediction Test	Submit multiple prediction requests simultaneously from different browser sessions.	The application handles multiple prediction requests without performance degradation.	Successfully processed concurrent requests.	Pass
PT-03	Dataset Loading Performance	Load the historical flood dataset during application startup and monitor processing time.	Dataset loads efficiently without application crash or excessive delay.	Dataset loaded successfully within acceptable time.	Pass